離散數學

Homework #1 (Due date: 10/12/2020)

1. (6 points) Which of these are propositions? What are the truth values of those that are propositions?
   a) Do not pass go.
   b) What time is it?
   c) There are no black flies in Maine.
   d) $4 + x = 5$.
   e) The moon is made of green cheese.
   f) $2^n \geq 100$.

   Propositions must have clearly defined truth values, so a proposition must be a declarative sentence with no free variables.
   a) This is not a proposition; it's a command.
   b) This is not a proposition; it's a question.
   c) This is a proposition that is false, as anyone who has been to Maine knows.
   d) This is not a proposition; its truth value depends on the value of x.
   e) This is a proposition that is false.
   f) This is not a proposition; its truth value depends on the value of n.

2. (6 points) Let p, q, and r be the propositions
   p :You get an A on the final exam.
   q :You do every exercise in this book.
   r :You get an A in this class.

   Write these propositions using p, q, and r and logical connectives (including negations).
   a) You get an A in this class, but you do not do every exercise in this book.
   b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
   c) To get an A in this class, it is necessary for you to get an A on the final.
   d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
   e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
   f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

   a) $r \wedge \neg q$ b) $p \wedge q \wedge r$ c) $r \rightarrow p$
   d) $p \wedge \neg q \wedge r$ e) $(p \wedge q) \rightarrow r$ f) $r \leftrightarrow (q \vee p)$

3. (4 points) For each of these sentences, determine whether an inclusive or, or an exclusive or, is intended. Explain your answer.
   a) Coffee or tea comes with dinner.
   b) A password must have at least three digits or be at least eight characters long.
   c) The prerequisite for the course is a course in number theory or a course in cryptography.
   d) You can pay using U.S. dollars or euros.

   a) exclusive or b) inclusive or
   c) inclusive or d) exclusive or

4. (8 points) Express these system specifications using the propositions p "The message is scanned for viruses" and q "The message was sent from an unknown system" together with logical connectives (including negations).

a) "The message is scanned for viruses whenever the message was sent from an unknown system."
b) "The message was sent from an unknown system but it was not scanned for viruses."
c) "It is necessary to scan the message for viruses whenever it was sent from an unknown system."
d) "When a message is not sent from an unknown system it is not scanned for viruses."

a) $q \rightarrow p$ b) $q \wedge \neg p$ c) $q \rightarrow p$ d) $\neg q \rightarrow \neg p$

5. (5 points) When planning a party you want to know whom to invite. Among the people you would like to invite are three touchy friends. You know that if Jasmine attends, she will become unhappy if Samir is there, Samir will attend only if Kanti will be there, and Kanti will not attend unless Jasmine also does. Which combinations of these three friends can you invite so as not to make someone unhappy?

We will translate these conditions into statements in symbolic logic, using j , s, and k for the propositions that Jasmine, Samir, and Kanti attend, respectively. The first statement is $j \rightarrow \neg s$. The second statement is $s \rightarrow k$. The last statement is $\neg k \vee j$, because "unless" means "or." (We could also translate this as $k \rightarrow j$. Also, we know that $p \rightarrow q$ is equivalent to q unless $\neg p$. In this case p is $\neg j$ and q is $\neg k$.) First, suppose that s is true. Then the second statement tells us that k is also true, and then the last statement forces j to be true. But now the first statement forces s to be false. So we conclude that s must be false; Samir cannot attend. On the other hand, if s is false, then the first two statements are automatically true, not matter what the truth values of k and j are. If we look at the last statement, we see that it will be true as long as it is not the case that k is true and j is false. So the only combinations of friends that make everybody happy are Jasmine and Kanti, or Jasmine alone (or no one!

6. (12 points) Show that each of these conditional statements is a tautology by using truth tables.

a) $[\neg p \wedge (p \vee q)] \rightarrow q$

b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

c) $[p \wedge (p \rightarrow q)] \rightarrow q$

d) $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

We construct a truth table for each conditional statement and note that the relevant column contains only T's. For part (a) we have the following table.

p	q	$\neg p$	$p \vee q$	$\neg p \wedge (p \vee q)$	$[\neg p \wedge (p \vee q)] \rightarrow q$
T	T	F	T	F	T
T	F	F	T	F	T
F	T	T	T	T	T
F	F	T	F	F	T

For part (b) we have the following table. We omit the columns showing $p \rightarrow q$ and $q \rightarrow r$ so that the table will fit on the page.

p	q	r	$(p \rightarrow q) \rightarrow (q \rightarrow r)$	$q \rightarrow r$	$[(p \rightarrow q) \rightarrow (q \rightarrow r)] \rightarrow (p \rightarrow r)$
T	T	T	T	T	T
T	T	F	F	T	T
T	F	T	T	T	F
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	F	T	F
F	F	T	T	T	F
F	F	F	T	T	T

For part (c) we have the following table.

p	q	$p \to q$	$p \wedge (p \to q)$	$[p \wedge (p \to q)] \to q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

For part (d) we have the following table. We have omitted some of the intermediate steps to make the table fit.

p	q	r	$(p \vee q) \wedge (p \to r) \wedge (p \to r)$	$[(p \vee q) \wedge (p \to r) \wedge (p \to r)] \to r$
T	T	T	T	T
T	T	F	F	T
T	F	T	T	T
T	F	F	F	T
F	T	T	T	T
F	T	F	F	T
F	F	T	F	T
F	F	F	F	T

7. (4 points) Find a compound proposition involving the propositional variables p, q, and r that is true when p and q are true and r is false, but is false otherwise. [Hint: Use a conjunction of each propositional variable or its negation.]

Following the hint, we easily see that the answer is $p \wedge q \wedge \neg r$.

8. (10 points) In recreational mathematics, a magic square is an n × n array of distinct positive integers in the range 1 , 2 , . . . , n × n such that the sums of the integers in each row, column and main diagonal (including the anti-diagonal) are equal. The integer n (where n is the number of integers on a side) is the order of the magic square and the constant sum is called the magic constant. Write a propositional formula f for a 4 × 4 magic square so that if an assignment for the atomic propositions makes f true then it corresponds to a solution for the 4 × 4 magic square puzzle.
Hint: consider the following atomic propositions:

P(i, j, k): the value of cell ij is k
R(i): the sum of row i is the magic constant
C(j): the sum of column j is the magic constant
D: the sum of the main diagonal (⋱) is the magic constant
A: the sum of the anti-diagonal (⋰) is the magic constant

$$f = \bigwedge_{k=1}^{16} \bigvee_{i=1}^{4} \bigvee_{j=1}^{4} P(i,j,k) \wedge \bigwedge_{i=1}^{4} R(i) \wedge \bigwedge_{j=1}^{4} C(j) \wedge D \wedge A$$

(By the way, it should be easy to extend the formula to n × n magic square.)

9. (8 points) Find a proposition function P(x, y) together with the domain for x and y so that $\forall x\, \exists y\, P(x,y)$,. $\exists y\, \forall x\, P(x,y)$, $\exists x\, \forall y\, P(x,y)$ are false and $\forall y \exists x\, P(x,y)$ is true.

Let a < b be two real numbers, and let U be {x| a ≤ x < b}. Define $P(x,y) : x > y$.

(There is no standard solution for this problem. Any reasonable solution will do.)

10. (8 points) Translate these statements into English, where $R(x)$ is "x is a rabbit" and $H(x)$ is "x hops" and the domain consists of all animals.

a) $\forall x(R(x) \rightarrow H(x))$ **b)** $\forall x(R(x) \wedge H(x))$

c) $\exists x(R(x) \rightarrow H(x))$ **d)** $\exists x(R(x) \wedge H(x))$

a) If an animal is a rabbit, then that animal hops. (Alternatively, every rabbit hops.)
b) Every animal is a rabbit and hops.
c) There exists an animal such that if it is a rabbit, then it hops.
d) There exists an animal that is a rabbit and hops. (Alternatively, some rabbits hop. Alternatively, some hopping animals are rabbits.)

Note that part (b) and part (c) are not the sorts of things one would normally say.

11. (6 points) Let $P(x)$ be the statement "$x = x^2$." If the domain consists of the integers, what are these truth values?

a) $P(0)$ **b)** $P(1)$ **c)** $P(2)$

d) $P(-1)$ **e)** $\exists x P(x)$ **f)** $\forall x P(x)$

a) T b) T c) F d) F e) T f) F

12. (6 points) Suppose that the domain of the propositional function $P(x)$ consists of the integers 0, 1, 2, 3, and 4. Write out each of these propositions using disjunctions, conjunctions, and negations.

a) $\exists x P(x)$ **b)** $\forall x P(x)$ **c)** $\exists x \neg P(x)$

d) $\forall x \neg P(x)$ **e)** $\neg \exists x P(x)$ **f)** $\neg \forall x P(x)$

a) $P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$
b) $P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4)$
c) $\neg P(0) \vee \neg P(1) \vee \neg P(2) \vee \neg P(3) \vee \neg P(4)$
d) $\neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$
e) $\neg(P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4))$
f) $\neg(P(0) \wedge P(1) \wedge P(2) \wedge P(3) \wedge P(4))$

13. (10 points) Define the following quantifiers:

a) $\exists_3 x$: $\exists_3 x P(x)$ means there are at least three different values for x such that P(x) is true.

b) $\exists_3! x$: $\exists_3! x\, P(x)$ means there are exactly three different values for x such that P(x) is true.

a) $\exists_3 x \equiv \exists x \exists y \exists z (P(x) \wedge P(y) \wedge P(z) \wedge x \neq y \wedge y \neq z \wedge z \neq x)$

b) $\exists_3! x \equiv \exists x \exists y \exists z (P(x) \wedge P(y) \wedge P(z) \wedge x \neq y \wedge y \neq z \wedge z \neq x \wedge \forall t (P(t) \rightarrow t=x \vee t=y \vee t=z))$

14. (10 points) Establish these logical equivalences, where x does not occur as a free variable in A. Assume that the domain is nonempty.

a) $\forall x(A \rightarrow P(x)) \equiv A \rightarrow \forall x P(x)$

b) $\exists x(A \rightarrow P(x)) \equiv A \rightarrow \exists x P(x)$

a) There are two cases. If A is false, then both sides of the equivalence are true, because a conditional statement with a false hypothesis is true. If A is true, then A $\rightarrow$ P(x) is equivalent to P(x) for each x, so the left-hand side is equivalent to $\forall x P(x)$, which is equivalent to the right-hand side.

b) There are two cases. If A is false, then both sides of the equivalence are true, because a conditional statement with a false hypothesis is true (and we are assuming that the domain is nonempty). If A is true, then A $\rightarrow$ P(x) is equivalent to P(x) for each x, so the left-hand side is equivalent to $\exists x P(x)$, which is equivalent to the right-hand side.

15. (12 points) Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

a) $\exists x \forall y(xy = y)$

b) $\forall x \forall y(((x < 0) \wedge (y < 0)) \rightarrow (xy > 0))$

c) $\exists x \exists y((x^2 > y) \wedge (x < y))$

d) $\forall x \forall y \exists z(x + y = z)$

a) There is a multiplicative identity for the real numbers.
b) The product of two negative real numbers is always a positive real number.
c) There exist real numbers x and y such that x2 exceeds y but x is less than y .
d) The real numbers are closed under the operation of addition.

16. (5 points) Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

a) $\neg \exists y \exists x P(x, y)$

b) $\neg \forall x \exists y P(x, y)$

c) $\neg \exists y(Q(y) \wedge \forall x \neg R(x, y))$

d) $\neg \exists y(\exists x R(x, y) \vee \forall x S(x, y))$

e) $\neg \exists y(\forall x \exists z\, T(x, y, z) \vee \exists x \forall z\, U(x, y, z))$

a) $\forall y \forall x \neg P(x, y)$
b) $\exists x \forall y\, \neg P(x, y)$
c) $\forall y(\neg Q(y) \vee \exists x R(x, y))$
d) $\forall y(\forall x \neg R(x, y) \wedge \exists x \neg S(x, y))$
e) $\forall y(\exists x \forall z\, \neg T(x, y, z) \wedge \forall x \exists z\, \neg U(x, y, z))$

17. (6 points) Determine the truth value of the statement $\exists x \forall y(x \leq y^2)$ if the domain for the variables consists of

a) the positive real numbers.

b) the integers.

c) the nonzero real numbers.

This statement says that there is a number that is less than or equal to all squares.
a) This is false.
b) This is true, since we can take x = −1, for example.
c) This is true, since we can take x = −1, for example.

18. (8 points) For each of these arguments, explain which rules of inference are used for each step.

a) "Linda, a student in this class, owns a red convertible. Everyone who owns a red convertible has gotten at least one speeding ticket. Therefore, someone in this class has gotten a speeding ticket."

b) "Each of five roommates, Melissa, Aaron, Ralph,Veneesha, and Keeshawn, has taken a course in discrete mathematics. Every student who has taken a course in discrete mathematics can take a course in algorithms. Therefore, all five roommates can take a course in algorithms next year."

In each case we set up the proof in two columns, with reasons, as in Example 6.

a) Let $c(x)$ be "x is in this class," let $r(x)$ be "x owns a red convertible," and let $t(x)$ be "x has gotten a speeding ticket." We are given premises $c(\text{Linda})$, $r(\text{Linda})$, $\forall x(r(x) \to t(x))$, and we want to conclude $\exists x(c(x) \wedge t(x))$.

Step	Reason
1. $\forall x(r(x) \to t(x))$	Hypothesis
2. $r(\text{Linda}) \to t(\text{Linda})$	Universal instantiation using (1)
3. $r(\text{Linda})$	Hypothesis
4. $t(\text{Linda})$	Modus ponens using (2) and (3)
5. $c(\text{Linda})$	Hypothesis
6. $c(\text{Linda}) \wedge t(\text{Linda})$	Conjunction using (4) and (5)
7. $\exists x(c(x) \wedge t(x))$	Existential generalization using (6)

b) Let $r(x)$ be "r is one of the five roommates listed," let $d(x)$ be "x has taken a course in discrete mathematics," and let $a(x)$ be "x can take a course in algorithms." We are given premises $\forall x(r(x) \to d(x))$ and $\forall x(d(x) \to a(x))$, and we want to conclude $\forall x(r(x) \to a(x))$. In what follows y represents an arbitrary person.

Step	Reason
1. $\forall x(r(x) \to d(x))$	Hypothesis
2. $r(y) \to d(y)$	Universal instantiation using (1)
3. $\forall x(d(x) \to a(x))$	Hypothesis
4. $d(y) \to a(y)$	Universal instantiation using (3)
5. $r(y) \to a(y)$	Hypothetical syllogism using (2) and (4)
6. $\forall x(r(x) \to a(x))$	Universal generalization using (5)

19. (6 points) Justify the rule of **universal transitivity**, which states that if $\forall x(P(x) \to Q(x))$ and $\forall x(Q(x) \to R(x))$ are true, then $\forall x(P(x) \to R(x))$ is true, where the domains of all quantifiers are the same.

We want to show that the conditional statement P(a) → R(a) is true for all a in the domain; the desired conclusion then follows by universal generalization. Thus we want to show that if P(a) is true for a particular a, then R(a) is also true. For such an a, by universal modus ponens from the first premise we have Q(a), and then by universal modus ponens from the second premise we have R(a), as desired.